

Mach-Zehnder Interferometer Proposal

Shannon Hamp
EdX Public Username: Shannon-Hamp
Spark Photonics Design
Waltham, Massachusetts 02451
Email: shamp@sparkphotonics.com

Abstract—The Mach-Zehnder interferometer (MZI) is a critical component of photonic integrated circuits that enables a wide range of functionality including optical switches, routers, and modulators. Here, I present the characterization and validation of MZIs with path length differences ranging from 50 to 200 μm . The MZIs were fabricated on a 220 nm silicon wafer for experimental verification of the simulated values for free spectral range and waveguide group index.

I. INTRODUCTION

Mach-Zehnder interferometers (MZIs) are optical devices that separate incident light between two paths and then recombine the two wavefronts, resulting in interference. In silicon photonics, MZIs are particularly relevant for the implementation of optical switches, where changes to the relative phase between the two interferometer arms produces constructive or destructive interference [1]. Therefore, control over the phase change between the two arms enables switching behavior at the output of the interferometer. This phase change can be implemented through physical differences in the path length between the two branches, thermo-optic control [2] by inducing a change in the refractive index of one arm by adding heat, as well as optical pumping [3]. Integrated photonic switching architectures can also be scaled up to enable larger switching circuits [4].

Here, I present the characterization and validation of MZIs with path length differences ranging from 50 to 200 μm . The MZI devices and loopback reference were developed to meet the design objective of enabling experimental measurement of the waveguide group index for comparison to simulated values.

II. THEORY

The output intensity of an MZI can be derived through the treatment of each passive optical element used to construct the interferometer. Considering the simple case of a y-splitter connected to a y-combiner (Fig. 1), the output electric field at each branch of the y-splitter can be related to the input electric field by Eq. 1, where E_i is the input field, E_1 is the output of branch 1, and E_2 is the output of branch 2.

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}} \quad (1)$$

Similarly, the input fields can be related to the output field of the y-combiner by Eq. 2, where E_1 and E_2 are the input fields at each branch, and E_o is the output field [1].

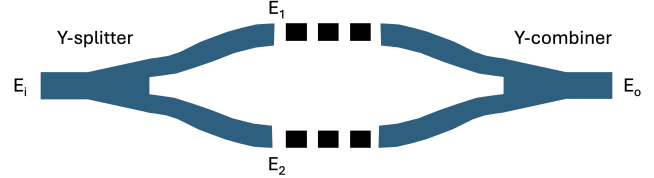


Fig. 1. Simple MZI schematic constructed with a y-splitter and y-combiner.

$$E_o = \frac{E_1 + E_2}{\sqrt{2}} \quad (2)$$

Then, connecting the y-splitter to the y-combiner produces an MZI such that for a plane wave described by $E = E_o e^{i(\omega t - \beta z)}$ at the input of the y-splitter, the field at the output of the y-combiner is given by Eq. 3, where E_i is the input field; E_o is the output field; β_1 and β_2 are the propagation constants for branch 1 and branch 2 of the splitter, respectively; α_1 and α_2 are the propagation losses for branch 1 and branch 2 of the splitter, respectively; L_1 and L_2 are the lengths of branch 1 and branch 2, respectively [1].

$$E_o = \frac{E_i}{2} (e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}) \quad (3)$$

To achieve a relationship in terms of measurable quantities, the electric fields were converted to intensity. Eq. 4 is an example of a lossless interferometer case, where the two waveguides were assumed to be the same such that the refractive indices in the two branches were equivalent ($\beta_1 = \beta_2$), I_i is the input intensity, I_o is the output intensity, and ΔL is the difference in path length between branch 1 and branch 2 [1].

$$I_o = \frac{I_i}{2} (1 + \cos(\beta \Delta L)) \quad (4)$$

From Eq. 4, the free spectral range (FSR) of an unbalanced MZI can be derived as the spacing between adjacent transmission peaks (Eq. 5), where λ is the operating wavelength and n_g is the group index, characterized by Eq. 6 [1].

$$FSR = \frac{\lambda^2}{\Delta L n_g} \quad (5)$$

$$n_g = n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda} \quad (6)$$

III. MODELING AND SIMULATION

The following section includes a discussion of waveguide models as well as interferometer design choices, models, and simulation results.

A. Waveguide

The MZI designs proposed here use TE silicon strip waveguides with a height of 220 nm as constrained by the silicon wafers used for fabrication [1]. Furthermore, the waveguide width for the primary set of MZIs was selected to be 500 nm. The electric field intensity corresponding to the TE mode in the waveguide with a 500 nm width is shown in Fig. 2.

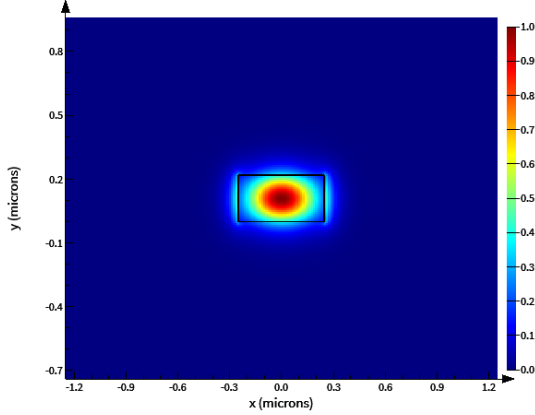


Fig. 2. Electric field intensity for a silicon strip waveguide with a height of 220 nm and a width of 500 nm.

Additional Lumerical MODE simulations were conducted for a wavelength sweep from 1500 nm to 1600 nm. The resulting simulation values for effective index and group index are included as a function of wavelength in Fig. 3.

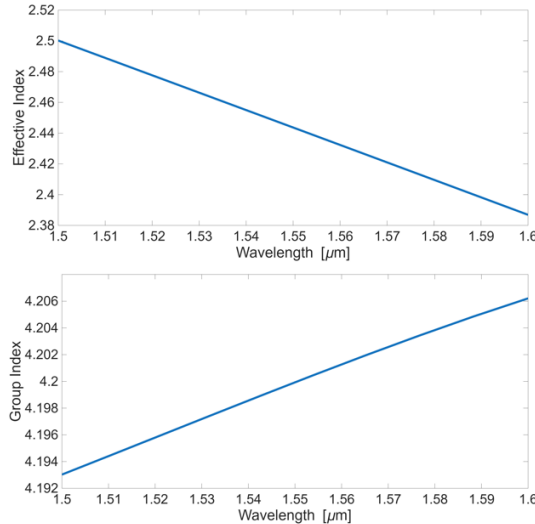


Fig. 3. Simulated effective index (top) and group index (bottom) for a silicon strip waveguide with a height of 220 nm and a width of 500 nm. Simulated values were acquired from Lumerical MODE across a frequency sweep from 1500 nm to 1600 nm.

The Lumerical MODE simulation effective index results were also used to develop a compact model for the effective index of the waveguide as a function of wavelength (Eq. 7). This was accomplished using a provided MATLAB script [1] to fit a third-order polynomial assuming a center wavelength of $\lambda_0 = 1550\text{nm}$. The resulting curve fit was plotted with the initial refractive index values from the MODE simulation (Fig. 4).

$$n_{eff}(\lambda) = 2.4437 - 1.1330(\lambda - \lambda_0) - 0.0431(\lambda - \lambda_0)^2 \quad (7)$$

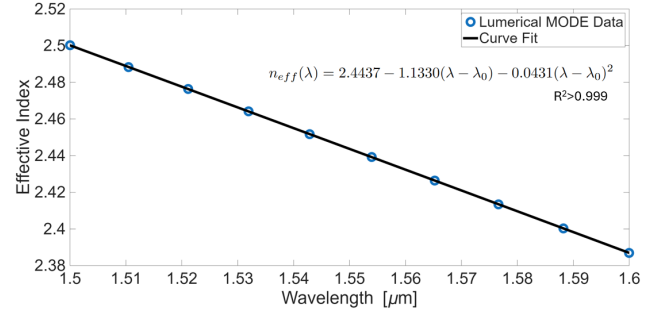


Fig. 4. Lumerical MODE simulation data for the proposed waveguide and corresponding compact model curve fit with an R-squared value greater than 0.999.

B. Mach-Zehnder Interferometer

The waveguide simulation results obtained by Lumerical MODE were then exported for further analysis within an MZI simulation in Lumerical INTERCONNECT. Six circuits—each with waveguide widths of 500 nm—were simulated in INTERCONNECT using y-splitters (Optical N Port S-Parameter), waveguides (MODE Waveguide with ldf file from MODE simulations discussed in previous section), and grating couplers (Optical S-Parameter with provided s-parameters file, "S_TE1550_SubGC_neg31_oxide.txt" [1]) connected as shown in Fig. 5

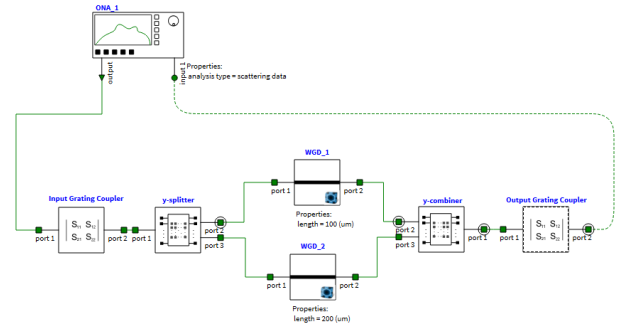


Fig. 5. Lumerical INTERCONNECT MZI schematic containing input and output grating couplers, along with two y-branches and two waveguides of different lengths.

The circuit model was then used to simulate the gain of the circuit (shown in Fig. 6 for path length differences of 100 and

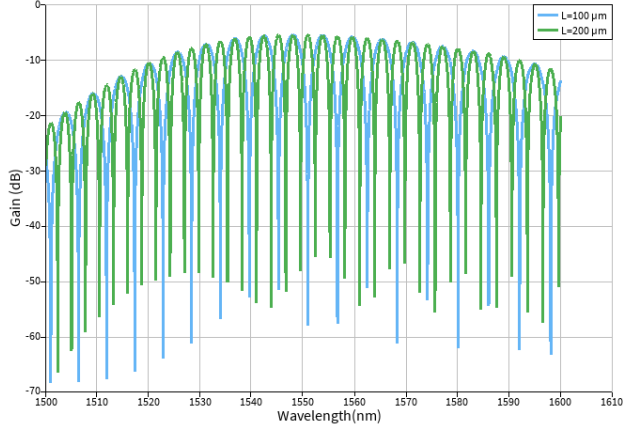


Fig. 6. Lumerical INTERCONNECT MZI simulation results for the gain (dB) for the circuits with 100 μm and 200 μm path length differences as a function of wavelength.

TABLE I
LUMERICAL INTERCONNECT SIMULATED FREE SPECTRAL RANGE VALUES FOR PATH LENGTH DIFFERENCES RANGING FROM 50 TO 200 μm .

ΔL [μm]	FSR [nm]
50	11.388
100	5.676
150	3.768
200	2.832

200 μm), as well as the free spectral range (FSR) with respect to varying path length differences (Table I).

The simulated transmission results for the 50 μm path length difference (Fig. 7) were validated with the first-order transfer function for the MZI (Eq. 8) derived from Eq. 4.

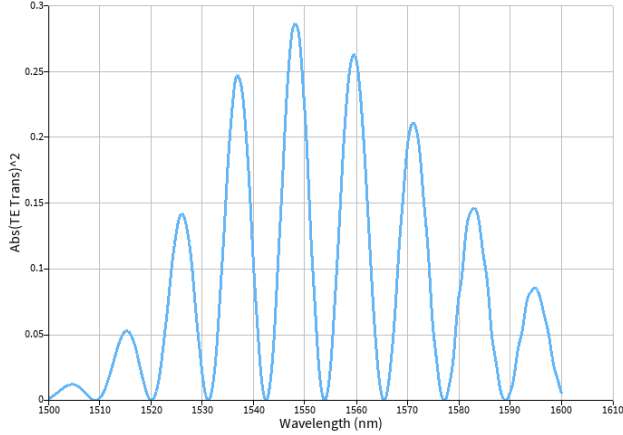


Fig. 7. Lumerical INTERCONNECT MZI simulation results for the transmission as a function of wavelength for the TE mode at a path length difference of 50 μm .

$$\frac{I_o}{I_i} = \frac{1}{2}(1 + \cos(\beta\Delta L)) \quad (8)$$

IV. FABRICATION

The layout in Fig. 8 was fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (ANT). The process relies on 100 keV electron beam lithography tools and uses silicon-on-insulator wafers with 200 mm diameter, 220 nm device thickness, and 2 μm buffer oxide thickness. The following description of the ANT process was quoted directly from the Summary of Fabrication from the Silicon Photonics Design, Fabrication, and Data Analysis edX module [5]:

The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H₂SO₄:H₂O₂) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

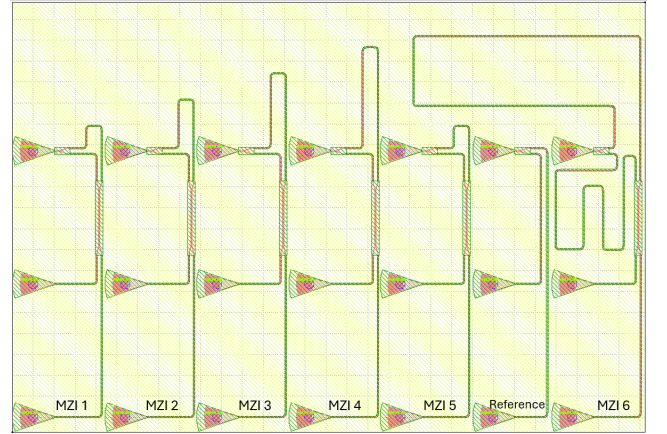


Fig. 8. Fabricated layout containing six MZIs with path lengths ranging from 50 to 200 μm , as well as a reference structure for calibration.

V. EXPERIMENT DATA

The devices in Fig. 8 were measured through the edX course. The following description of the measurement process was quoted directly from the Measurement Description from the Silicon Photonics Design, Fabrication, and Data Analysis edX module [5]:

The devices were characterized using a custom-built automated test setup [1], [6] with automated control software written in Python was used [7]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [8]. A 90° rotation was used to inject light into the TM grating couplers [8]. A polarization maintaining fibre array was used to couple light in/out of the chip [9].

VI. ANALYSIS

The raw measurement data for the six MZIs Fig. 8 were calibrated to remove spectral structure due to the coupling efficiency of the grating couplers using the reference device in the layout. The calibrated data were then fit with a polynomial to empirically calibrate the spectrum by flattening any remaining spectral amplitude shifts. Then, the wavelength range of the measurement data was reduced to limit noisy data in following analysis. The FSR and group index were then computed using the peaks of spectrum that was created using an autocorrelation-based frequency estimation of the calibrated spectrum (Fig. 9). These processing steps were completed for each MZI, and the experimental FSR and group indices were reported in Tables II and III were the average of the channel 1 results and the channel 2 results measured from the first and second output grating couplers.

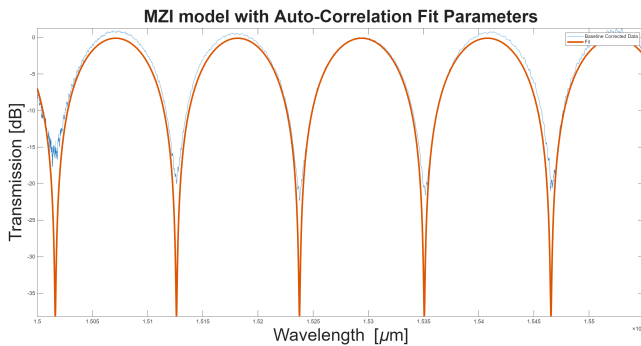


Fig. 9. Transmission spectrum for MZI1 (path length of 50 μm) including the baseline-corrected measurement data and the fit used to calculate FSR and group index.

The percent difference between simulated group indices and experimental group indices in Table III were all less than 1.5%, demonstrating that all MZI devices performed in alignment with simulated values. Additionally, the experimental group index of MZI1, 2, 3, and 6 all fell within the minimum

TABLE II
THEORETICAL AND EXPERIMENTAL VALUES FOR THE FSR OF THE 6 MZI DEVICES.

Circuit	ΔL [μm]	Sim FSR [nm]	Exp FSR [nm]	Diff [%]
MZI1	50	11.41	11.388	0.193
MZI2	100	5.62	5.676	-0.996
MZI3	150	3.79	3.768	0.580
MZI4	200	2.839	2.832	0.247
MZI5	50	11.41	11.344	0.578
MZI6	200	2.851	2.836	0.526

TABLE III
THEORETICAL AND EXPERIMENTAL VALUES FOR THE WAVEGUIDE GROUP INDEX OF THE 6 MZI DEVICES.

Circuit	ΔL [μm]	Sim n_g [nm]	Exp n_g [nm]	Diff [%]
MZI1	50	4.199	4.219	-0.485
MZI2	100	4.199	4.233	-0.803
MZI3	150	4.199	4.251	-1.231
MZI4	200	4.199	4.242	-1.017
MZI5	50	4.199	4.236	-0.874
MZI6	200	4.199	4.236	-0.874

and maximum corner analysis values of 4.1587 and 4.2369, respectively.

VII. CONCLUSION

The MZI designs presented here were fabricated, measured, and calibrated to experimentally determine the waveguide group index. Four of the six MZI designs produced experimental waveguide group indices within the minimum and maximum simulated corner analysis values for the group index. Future work to further characterize the waveguide group index could include additional MZIs with a wider range of path length difference and multiple calibration structures across the chip the better account for spatial variation on the chip.

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